

Tae Yong Song

Ph.D. Student in Building Science

Department of Architecture and Architectural Engineering, Seoul National University

Website | Google Scholar | GitHub | LinkedIn

EDUCATION

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|--------------|---|
| 2026–present | Ph.D. in Architecture and Architectural Engineering , Seoul National University

Building science and building performance simulation |
| 2022–2023 | M.S. in Architecture and Architectural Engineering , Seoul National University

Thesis: “Exploring Potential of Transfer Learning in Overcoming Data Scarcity for Building Energy Model” |
| 2015–2021 | B.S. in Architecture and Architectural Engineering , Seoul National University |

PUBLICATIONS

International Journal Article

- 2025** Physics-embedded hybrid modelling approach for room temperature prediction using Siamese neural network and RC model

Chul-Hong Park, Seongkwon Cho, **Tae Yong Song**, Seon-Young Heo, Junu Lee, and Cheol-Soo Park. *Journal of Building Performance Simulation*, 19(2), 277–288. Published online 5 August 2025; issue dated 2026.

International Conference Papers

- 2026** Curated Intelligence for Visual Inference and Deterministic Logic toward Automatic BEM

Tae Yong Song, Jin-Hong Kim, Seung-Ju Lee, Soo-Kyung Kim, Hyun-Chul Park, Kyung-Jae Kim, and Cheol-Soo Park. *Proceedings of the 7th Building Simulation Applications Conference*, June 24–26, Bozen-Bolzano, Italy.
- 2024** Physics-Informed Hybrid Modeling Approach for Room Temperature Prediction Using an RC Model and Siamese Neural Network

Chul-Hong Park, Seongkwon Cho, **Tae Yong Song**, Seon-Young Heo, and Cheol-Soo Park. *SimBuild 2024*.
- 2023** Causality-based model for decision-making of building energy retrofit

Chul-Hong Park, Young-Sub Kim, **Tae Yong Song**, Seon-Jung Ra, and Cheol-Soo Park. *Building Simulation 2023*, 1433–1438.
- 2023** Limitations of Conventional Artificial Neural Network-Based Surrogate Models for Building Energy Retrofit

Chul-Hong Park, Young-Sub Kim, Seon-Jung Ra, **Tae Yong Song**, and Cheol-Soo Park. *ASHRAE Annual Conference*, 683–690.

Domestic Conference Papers

- 2026** Development of Customized AI Technologies for BIM-to-BEM Automation: Integration of LLMs, Computer Vision, Delaunay Geometry Algorithms, and Minimum Cycle Algorithms

Tae Yong Song, Jin-Hong Kim, Soo-Kyung Kim, Hyun-Chul Park, Kyung-Jae Kim, and Cheol-Soo Park. *Proceedings of the 2026 Spring Annual Conference of the Architectural Institute of Korea*, 46(1), 829–830. **Awarded the Outstanding Presentation Paper Award (General Division, Oral Presentation).**
- 2022** Dynamic Sensitivity Analysis Using NODE-GAM

Tae Yong Song, Chul-Hong Park, Seon-Jung Ra, Young-Sub Kim, and Cheol-Soo Park. *Proceedings of the 2022 Annual Conference of the Korean Institute of Architectural Sustainable Environment and Building Systems*, 181–182.

RESEARCH PROJECTS

2026–2030	Virtual Building Modeling and Simulation Framework, and a Causality-Based AR Tool for Net-Zero Building Design Evaluation
2025–2026	LLM Technology for Digital-Twin-Based Building Energy Models
2025–2026	IDF Model Development for Digital-Twin Energy Simulation
2022–2024	Automated Digital Diagnosis and Design Technologies for Green Remodeling
2023	Data-Driven HVAC Load Prediction and Optimal Air-Handling-Unit Control
2022–2023	Greenhouse-Gas Reduction Factors and Emissions Assessment for Building Green Remodeling

TEACHING EXPERIENCE

2026–present	Teaching Assistant, Seoul National University Department of Architecture and Architectural Engineering
2023	Teaching Assistant, Seoul National University Department of Architecture and Architectural Engineering

HONOR

2026	Outstanding Presentation Paper Award Architectural Institute of Korea Spring Annual Conference, Oral Presentation – General Division
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